

Latent Dense Associative Memory Force Matching for Physical Score-Based Models

Analysis, learning rules, expressivity, and crossbar implementation

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Scope. This note analyzes whether the local, on-device force-matching framework for score-based generative models in analogue oscillator networks can be extended from visible-only energies to a restricted latent architecture in which hidden units equilibrate on a fast time scale. It uses the diffusion/Dense Associative Memory (DenseAM) correspondence and derives marginal/free-energy learning rules for hidden-unit models. The emphasis is on a physical architecture with full-amplitude visible clamping, fast hidden equilibration, and trainable visible-hidden crossbar couplings.

Bottom line. The idea is technically coherent and potentially strong. The clean formulation is not to score-match a joint visible-hidden density, but to score-match the *visible marginal* obtained after adiabatically eliminating the hidden variables. This turns the hidden layer into a fast free-energy module. In a restricted Boltzmann/DenseAM architecture the resulting force and the main denoising force-matching update are local or column-local, and map naturally to crossbar operations. The major technical risks are hidden-temperature calibration, finite time-scale separation, nonideal crossbar reciprocity, and avoiding a regime where extra capacity simply memorizes.

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1 Executive assessment

1.1 Answers to the four concrete questions

1. Can the local on-device training rule be extended to hidden units using DenseAM concepts? Yes, provided the hidden units are treated as latent variables that are equilibrated or approximately equilibrated under a visible clamp. The correct object is the visible free energy

$$\mathcal{F}_\theta(x, t) = -\beta^{-1} \log \int \exp[-\beta E_\theta(x, h, t)] dh, \quad \beta = (k_B T)^{-1}, \quad (1)$$

not the bare joint energy. The model score is

$$s_\theta(x, t) = \nabla_x \log p_\theta(x, t) = -\beta \nabla_x \mathcal{F}_\theta(x, t). \quad (2)$$

Training still uses a full-amplitude clamp of the visible state x_t to a noisy data sample. The hidden layer is not clamped; it relaxes rapidly to $q_\theta(h | x_t, t)$. The measured visible force is then the marginal force

$$f_\theta(x, t) = -\nabla_x \mathcal{F}_\theta(x, t) = -\langle \nabla_x E_\theta(x, h, t) \rangle_{q_\theta(h | x, t)}. \quad (3)$$

This is the same score-based physical principle as the visible-only model, but with an entropic hidden contribution. For a one-layer restricted architecture, the needed hidden statistics are means and susceptibilities. No infinitesimal visible nudging is required.

2. Would this improve expressivity, and to what extent? Yes. A fast hidden layer converts a visible-only polynomial/pairwise energy into a free energy containing log-sum-exp or softplus nonlinearities. Those nonlinearities induce high-order visible interactions and global mixture-like basins without explicitly wiring all high-order visible couplings. In the categorical case, the hidden layer is exactly the latent memory index of a DenseAM/log-sum-exp energy. In the binary hidden case, the score has the form of a tied autoencoder/RBM force field. With m hidden units and d visible units, the hardware cost is roughly $O(dm)$ couplings, while the induced visible free energy can represent many high-order interactions. The practical gain is largest when the current visible-only oscillator network is constrained by local coupling topology or low-order polynomial forces. The caveat is that increased capacity can move the system toward memorization unless capacity, temperature, sparsity, time dependence, and hidden-unit count are controlled.

3. Is this implementable in crossbar arrays? The core multiply-and-accumulate operations are very well matched to crossbars: hidden fields use $W^\top x$, visible forces use Wm , and the trainable parameters are conductances in a visible-hidden bipartite array. A physically conservative energy requires reciprocal/tied couplings, or a calibrated pair of arrays. Signed weights require differential conductance pairs or offset encodings. The exact free-energy learning rule requires either hidden susceptibilities or a denoising target-score error. These are feasible but not automatic; they require column-level aggregates, local update circuitry, and careful noise/temperature calibration. The idea is more plausible as a restricted one-hidden-layer architecture than as a deep/hierarchical DAM.

4. Is it new and potentially powerful? The components are not individually new: score matching for latent energy-based models, Gaussian-binary RBM free energies, DenseAM/diffusion equivalences, and analog/crossbar associative memories all exist. The specific synthesis – *local on-device score/force matching of a fast-equilibrating latent DenseAM/RBM free energy inside an analogue diffusion sampler, with full-amplitude visible clamping and trainable crossbar couplings* – appears novel relative to the literature I found. It is a credible research direction because it combines a real expressivity upgrade with an architecture that maps to physical hardware. It is not a guaranteed win; the main question is whether exact-enough hidden equilibration and susceptibility/error signals can be obtained without destroying the simplicity of the original force-matching proposal.

1.2 Recommended minimal proposal

Start with a one-hidden-layer restricted model:

$$E_{\theta}(x, h, t) = U_v(x, t) + U_h(h, t) - \sum_{i=1}^d \sum_{a=1}^m W_{ia}(t) x_i h_a, \quad (4)$$

where x are visible oscillator amplitudes and h are fast hidden units. Train only $W(t)$ and, if hardware permits, hidden biases $b_a(t)$. Avoid hidden-hidden and visible-visible learned couplings in the first implementation. Use denoising force matching for the first experiments, because it avoids the Laplacian/free-energy Hessian term and gives the cleanest physical update. Then compare against the self-contained Hyvarinen score-matching rule.

2 Why DenseAM suggests the right hidden-variable architecture

The key observation behind the diffusion/DenseAM correspondence is that the empirical noised data distribution is a mixture of kernels. In a variance-exploding Gaussian case with training patterns $\{\xi^{\mu}\}_{\mu=1}^K$, the marginal density at noise scale σ_t is

$$p_t(x) = \frac{1}{K} \sum_{\mu=1}^K \frac{1}{(2\pi\sigma_t^2)^{d/2}} \exp \left[-\frac{\|x - \xi^{\mu}\|^2}{2\sigma_t^2} \right]. \quad (5)$$

Up to constants, its negative log-density is

$$E_{\text{DM}}(x, t) = -2\sigma_t^2 \log \sum_{\mu=1}^K \exp \left[-\frac{\|x - \xi^{\mu}\|^2}{2\sigma_t^2} \right], \quad (6)$$

which is a log-sum-exp DenseAM energy. This energy can be interpreted as the free energy obtained by marginalizing a fast categorical latent memory index μ . In other words, the “hidden variable” view is not an arbitrary extension of DenseAM; it is already implicit in the log-sum-exp energy.

For a categorical hidden state $h = \mu$ with energy

$$E(x, \mu, t) = \frac{1}{2\sigma_t^2} \|x - \xi^{\mu}\|^2 - b_{\mu}(t), \quad (7)$$

the hidden posterior responsibility is

$$r_{\mu}(x, t) = q(\mu | x, t) = \frac{\exp[-\beta E(x, \mu, t)]}{\sum_{\nu} \exp[-\beta E(x, \nu, t)]}. \quad (8)$$

The marginal force is then a responsibility-weighted attraction toward memories:

$$-\nabla_x \mathcal{F}(x, t) = \sum_{\mu} r_{\mu}(x, t) \frac{\xi^{\mu} - x}{\sigma_t^2}. \quad (9)$$

This is precisely the kind of fast hidden equilibration the proposed physical system can exploit: visible units are clamped or evolving slowly, while hidden units compute posterior responsibilities or hidden activations.

3 Latent free-energy formulation

3.1 Adiabatic hidden equilibration

Let $x \in \mathbb{R}^d$ denote visible coordinates and $h \in \mathcal{H}$ denote hidden coordinates. At diffusion time t , define a joint physical energy $E_\theta(x, h, t)$ and assume hidden units thermalize on time scale τ_h much faster than the visible time scale τ_x :

$$\tau_h \ll \tau_x. \quad (10)$$

For fixed visible x , hidden variables follow

$$q_\theta(h | x, t) = \frac{\exp[-\beta E_\theta(x, h, t)]}{Z_h(x, t)}, \quad Z_h(x, t) = \int \exp[-\beta E_\theta(x, h, t)] dh. \quad (11)$$

Adiabatic elimination gives the visible free energy

$$\mathcal{F}_\theta(x, t) = -\beta^{-1} \log Z_h(x, t). \quad (12)$$

The visible marginal model is

$$p_\theta(x, t) \propto \exp[-\beta \mathcal{F}_\theta(x, t)]. \quad (13)$$

Thus the score learned by the physical system is

$$s_\theta(x, t) = -\beta \nabla_x \mathcal{F}_\theta(x, t). \quad (14)$$

The useful identities are

$$\nabla_x \mathcal{F}_\theta(x, t) = \langle \nabla_x E_\theta(x, h, t) \rangle_{q(h|x, t)}, \quad (15)$$

$$\nabla_x^2 \mathcal{F}_\theta(x, t) = \langle \nabla_x^2 E_\theta \rangle_q - \beta \text{Cov}_q(\nabla_x E_\theta, \nabla_x E_\theta), \quad (16)$$

$$\partial_{\theta_i} \mathcal{F}_\theta(x, t) = \langle \partial_{\theta_i} E_\theta(x, h, t) \rangle_q. \quad (17)$$

Equation (15) is the physical force identity: when the visible state is clamped, the force measured at the visible coordinate after hidden equilibration is the gradient of the free energy, not just the gradient of a bare energy at a fixed hidden state.

3.2 Reverse-time physical sampler

The visible-only oscillator proposal uses an energy $E_\theta^b(x, t)$ such that

$$-\frac{1}{k_B T} \nabla_x E_\theta^b(x, t) \approx \nabla_x \log p_t(x). \quad (18)$$

With hidden units, replace E_θ^b by the free energy $\mathcal{F}_\theta$:

$$-\frac{1}{k_B T} \nabla_x \mathcal{F}_\theta(x, t) \approx \nabla_x \log p_t(x). \quad (19)$$

For the Ornstein-Uhlenbeck forward process used in the oscillator paper, the reverse physical visible dynamics become

$$dx = [-2\nabla_x \mathcal{F}_\theta(x, t) + x] dt + \sqrt{2k_B T} dw, \quad (20)$$

with hidden variables either continuously equilibrated or implemented by a fast deterministic/stochastic circuit that realizes $q_\theta(h | x, t)$ or its conditional mean.

4 Learning rules with latent units

4.1 Exact marginal score matching

The Hyvarinen score-matching objective for the visible marginal at a fixed diffusion time is

$$J_{\text{SM}}(\theta, t) = \mathbb{E}_{x \sim p_t} \left[\frac{1}{2} \|s_\theta(x, t)\|^2 + \nabla_x \cdot s_\theta(x, t) \right]. \quad (21)$$

Substituting $s_\theta = -\beta \nabla_x \mathcal{F}_\theta$ gives

$$J_{\text{SM}}(\theta, t) = \mathbb{E}_{x \sim p_t} \left[-\beta \text{Tr} \nabla_x^2 \mathcal{F}_\theta(x, t) + \frac{\beta^2}{2} \|\nabla_x \mathcal{F}_\theta(x, t)\|^2 \right]. \quad (22)$$

Differentiating with respect to parameter θ_i yields the exact latent analogue of the visible-only force-matching rule:

$$\frac{\partial J_{\text{SM}}}{\partial \theta_i} = \mathbb{E}_{x \sim p_t} \left[-\beta \text{Tr} \nabla_x^2 \partial_{\theta_i} \mathcal{F}_\theta(x, t) + \beta^2 \nabla_x \mathcal{F}_\theta(x, t)^\top \nabla_x \partial_{\theta_i} \mathcal{F}_\theta(x, t) \right]. \quad (23)$$

This has the same structure as the visible-only rule, with E_θ^b replaced by $\mathcal{F}_\theta$. The difference is that $\partial_{\theta_i} \mathcal{F}_\theta$ is a hidden-equilibrium average. In general,

$$\nabla_x \partial_{\theta_i} \mathcal{F}_\theta = \langle \nabla_x \partial_{\theta_i} E_\theta \rangle_q - \beta \text{Cov}_q(\partial_{\theta_i} E_\theta, \nabla_x E_\theta). \quad (24)$$

Thus the exact rule requires hidden conditional means and covariances. In a restricted architecture these reduce to local activations and susceptibilities.

4.2 Denoising force matching: the cleanest on-device rule

For physical implementation, a denoising version is especially attractive. Suppose the forward corruption has conditional density $q_t(x | y)$ and known conditional score

$$r_t(x, y) = \nabla_x \log q_t(x | y). \quad (25)$$

For the OU forward process $x_t = a_t y + \sigma_t \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$, one has

$$r_t(x_t, y) = -\frac{x_t - a_t y}{\sigma_t^2}. \quad (26)$$

Define the denoising score loss

$$L_{\text{DSM}}(\theta, t) = \frac{1}{2} \mathbb{E}_{y, \epsilon} \|s_\theta(x_t, t) - r_t(x_t, y)\|^2. \quad (27)$$

Let

$$\delta(x_t, y, t) = r_t(x_t, y) - s_\theta(x_t, t) \quad (28)$$

be the target-minus-model score error. Since $s_\theta = -\beta \nabla_x \mathcal{F}_\theta$, gradient descent gives

$$\Delta \theta_i = -\eta \frac{\partial L_{\text{DSM}}}{\partial \theta_i} = -\eta \beta \mathbb{E} \left[\delta(x_t, y, t)^\top \nabla_x \partial_{\theta_i} \mathcal{F}_\theta(x_t, t) \right]. \quad (29)$$

The minus sign in Eq. (29) is important: in the common restricted architectures below, $\nabla_x \partial_{\theta_i} \mathcal{F}$ is negative for an attractive visible-hidden coupling, so the update becomes Hebbian-like with a target-score error.

This rule is often the most useful proposal experimentally:

- It uses a full-amplitude visible clamp to x_t .
- Hidden units equilibrate under that clamp.
- It does not need an infinitesimal visible perturbation.
- It avoids estimating the Laplacian term in Eq. (22).
- It requires a local score-error signal, which is available in synthetic diffusion training because the corruption noise is known.

5 Restricted binary hidden layer: explicit local rule

5.1 Energy and free energy

Consider binary hidden units $h_a \in \{0, 1\}$ with no hidden-hidden couplings:

$$E_\theta(x, h, t) = U_v(x, t) - \sum_{a=1}^m h_a \left[b_a(t) + \sum_{i=1}^d W_{ia}(t) x_i \right]. \quad (30)$$

Let

$$I_a(x, t) = b_a(t) + \sum_i W_{ia}(t) x_i, \quad m_a(x, t) = \langle h_a \rangle = \text{sigmoid}(\beta I_a). \quad (31)$$

The visible free energy is

$$\mathcal{F}_\theta(x, t) = U_v(x, t) - \beta^{-1} \sum_{a=1}^m \log(1 + \exp[\beta I_a(x, t)]). \quad (32)$$

Its gradient and score are

$$\nabla_x \mathcal{F}_\theta(x, t) = \nabla_x U_v(x, t) - W(t) m(x, t), \quad (33)$$

$$s_\theta(x, t) = \beta [W(t) m(x, t) - \nabla_x U_v(x, t)]. \quad (34)$$

Thus the hidden layer acts like a fast nonlinear encoder $m = \text{sigmoid}(\beta W^\top x + \beta b)$ and a tied decoder Wm that supplies the visible force.

Define the hidden susceptibility

$$\chi_a(x, t) = \frac{\partial m_a}{\partial I_a} = \beta m_a(1 - m_a). \quad (35)$$

For a coupling W_{ka} ,

$$\partial_{W_{ka}} \mathcal{F}_\theta(x, t) = -x_k m_a, \quad (36)$$

and

$$\frac{\partial}{\partial x_i} \partial_{W_{ka}} \mathcal{F}_\theta(x, t) = -\delta_{ik} m_a - x_k \chi_a W_{ia}. \quad (37)$$

Substituting Eq. (37) into Eq. (29) gives the denoising force-matching update

$$\Delta W_{ka}(t) = \eta \beta \left[m_a \delta_k + x_k \chi_a \sum_i W_{ia} \delta_i \right]. \quad (38)$$

Similarly, for hidden bias b_a ,

$$\Delta b_a(t) = \eta \beta \chi_a \sum_i W_{ia} \delta_i. \quad (39)$$

5.2 Interpretation of the binary rule

The first term in Eq. (38), $m_a \delta_k$, is the direct local Hebbian/error term: visible score error at unit k times hidden activation a . The second term, $x_k \chi_a (W_{:a}^\top \delta)$, corrects for the fact that changing W_{ka} also changes the hidden posterior $m_a(x)$. This term is a column-local susceptibility term. A crossbar column can naturally compute $W_{:a}^\top \delta$ as a current; x_k , m_a , and χ_a are locally available or broadcast along row/column lines.

A simplified hardware rule that drops the susceptibility correction,

$$\Delta W_{ka} \approx \eta \beta m_a \delta_k, \quad (40)$$

is a biased but plausible first-order “frozen hidden” approximation. It may work when hidden susceptibilities are small, when hidden activations are saturated, or as a first prototype. The exact free-energy gradient requires Eq. (38).

6 Categorical hidden layer: the DenseAM memory index

A categorical hidden variable $a \in \{1, \dots, K\}$ gives a direct DenseAM/log-sum-exp energy:

$$E_\theta(x, a, t) = U_v(x, t) - b_a(t) - w_a(t)^\top x. \quad (41)$$

Then

$$m_a(x, t) = q(a | x, t) = \text{softmax}_a \left(\beta [b_a + w_a^\top x] \right), \quad (42)$$

and

$$\mathcal{F}_\theta(x, t) = U_v(x, t) - \beta^{-1} \log \sum_a \exp \{ \beta [b_a + w_a^\top x] \}. \quad (43)$$

The force is

$$-\nabla_x \mathcal{F}_\theta(x, t) = -\nabla_x U_v(x, t) + \sum_a m_a(x, t) w_a(t). \quad (44)$$

With $U_v(x) = \frac{1}{2} \|x\|^2$, this is a modern Hopfield/DenseAM update force: the system is pulled toward the posterior-weighted memory vector. The denoising update for memory weight w_{ka} is

$$\boxed{\Delta w_{ka} = \eta \beta \left[m_a \delta_k + x_k \beta m_a (w_a - \bar{w})^\top \delta \right], \quad \bar{w} = \sum_b m_b w_b.} \quad (45)$$

The update for b_a is

$$\Delta b_a = \eta \beta^2 m_a (w_a - \bar{w})^\top \delta. \quad (46)$$

The categorical hidden architecture is the most literal DAM implementation. Its cost is one hidden unit per memory/prototype. The binary hidden architecture is more RBM-like and can represent combinations of features through multiple active hidden units.

7 Self-contained Hyvarinen rule for the binary restricted model

If one wants to stay maximally close to the visible-only score-matching rule and avoid an explicit denoising target, the restricted binary model admits a compact objective. Let

$$F_i(x) = \partial_{x_i} \mathcal{F}(x) = \partial_{x_i} U_v(x) - \sum_a W_{ia} m_a, \quad (47)$$

with $\chi_a = \beta m_a(1 - m_a)$. The trace of the Hessian is

$$\text{Tr} \nabla_x^2 \mathcal{F}(x) = \text{Tr} \nabla_x^2 U_v(x) - \sum_a \chi_a \|w_a\|^2. \quad (48)$$

Thus

$$J_{\text{SM}} = \mathbb{E}_x \left[-\beta \text{Tr} \nabla_x^2 U_v(x) + \beta \sum_a \chi_a \|w_a\|^2 + \frac{\beta^2}{2} \|\nabla_x U_v(x) - Wm(x)\|^2 \right]. \quad (49)$$

For W_{ka} , define $\chi'_a = \partial \chi_a / \partial I_a = \beta \chi_a(1 - 2m_a)$. Then

$$\frac{\partial J_{\text{SM}}}{\partial W_{ka}} = \mathbb{E}_x \left[-\beta^2 \left(m_a F_k + x_k \chi_a w_a^\top F \right) + \beta \left(2\chi_a W_{ka} + x_k \chi'_a \|w_a\|^2 \right) \right]. \quad (50)$$

Gradient descent uses $\Delta W_{ka} = -\eta \partial J_{\text{SM}} / \partial W_{ka}$. This is still local/column-local: it requires the measured visible force $-F_k$, hidden activation m_a , hidden susceptibility χ_a , a column aggregate $w_a^\top F$, and the column norm $\|w_a\|^2$.

Compared with Eq. (38), Eq. (50) is more self-contained because it does not require a target score error. It is less convenient in hardware because it needs the trace/susceptibility regularizer. The two should agree in the standard denoising-score-matching limit when the denoising objective is averaged over the forward corruption distribution.

8 Physical training protocol

A concrete on-device training cycle for the denoising rule is:

1. Choose a diffusion time t_ℓ and draw a data sample y .
2. Generate the noisy visible clamp $x_t = a_t y + \sigma_t \epsilon$ using known injected noise.
3. Clamp visible oscillators to x_t at full amplitude. Do not clamp hidden units.
4. Let hidden units equilibrate on the fast time scale under the current $W(t_\ell)$ and $b(t_\ell)$.
5. Measure or compute hidden means m_a , susceptibilities χ_a , and the effective visible force $f_i = -\partial_{x_i} \mathcal{F}$.
6. Compute the model score $s_i = \beta f_i$ and the target score $r_i = -(x_{t,i} - a_t y_i) / \sigma_t^2$.
7. Form $\delta_i = r_i - s_i$.
8. Apply local or column-local update pulses according to Eq. (38) or Eq. (45).
9. Iterate over samples and time slices. As in the visible-only protocol, initialize the optimization at $t_{\ell+1}$ from the learned parameters at t_ℓ to encourage a smooth driving protocol.

At inference, initialize x from the Gaussian prior, then run the reverse-time visible dynamics while hidden units continuously track the current visible state. If $W(t)$ is expensive to vary continuously in hardware, use a low-rank time basis

$$W(t) = \sum_{r=1}^R a_r(t) W^{(r)} \quad (51)$$

with a small number of physical arrays and programmable gains, or train a fixed W with time-dependent hidden biases/gains.

9 Expressivity analysis

9.1 What hidden units buy over visible-only oscillators

A visible-only oscillator energy built from low-order local and pairwise polynomial terms has to represent all global structure through explicit visible-visible couplings. A restricted hidden layer changes this qualitatively. For binary hidden units, the free energy contains softplus terms

$$-\beta^{-1} \log(1 + \exp[\beta(b_a + w_a^\top x)]). \quad (52)$$

Expanding such terms around an operating point produces mixed interactions of all orders in the visible coordinates. One hidden unit contributes a ridge-like nonlinear feature; many hidden units tile the data manifold with nonlinear basins. The score

$$s(x) = \beta(Wm(x) - \nabla U_v(x)) \quad (53)$$

is a tied-weight one-hidden-layer neural vector field constrained to be conservative because it is the gradient of a free energy.

For categorical hidden units, the log-sum-exp free energy creates a soft competition between memory/prototype vectors. This is exactly the mathematical structure that links DenseAMs to diffusion energies. The expressivity gain is therefore not just “more parameters”; it is a change in the functional class from explicit low-order potentials to marginal free energies with sharp, temperature-controlled attractor basins.

9.2 How much expressivity?

There are three useful regimes:

1. **Categorical DenseAM regime.** With K hidden memory units and $U_v(x) = \|x\|^2/2$, the model represents a log-sum-exp over K prototypes. At high inverse temperature, it creates sharp basins around prototypes; at lower inverse temperature, it interpolates. This is close to an empirical kernel density/diffusion energy. It is expressive but memory-heavy.
2. **Binary RBM regime.** With m binary hidden units, the visible marginal contains m softplus factors. The number of latent configurations is 2^m , although not all are independent modes. This gives a combinatorial feature representation with only $O(dm)$ crossbar weights.
3. **Compressed DAM/random-feature regime.** Hidden units can represent random or learned features rather than one hidden unit per training memory. This is likely the best route for hardware if the number of data points is large.

The practical expectation is that a moderate hidden layer should improve sample quality and mode coverage relative to visible-only local couplings, especially on data with global correlations. The largest risk is memorization: if m or K is too large relative to data size and noise temperature, the model may return to a memory-retrieval regime rather than a manifold/generalization regime. This is not a reason to reject the idea; it suggests that DenseAM diagnostics such as nearest-neighbor memorization, spurious-state fraction, and basin-volume measurements should be part of the evaluation.

10 Crossbar implementation assessment

10.1 Core circuit mapping

The restricted model maps naturally to crossbar operations:

Mathematical quantity	Operation	Physical realization
Hidden field $I_a = b_a + \sum_i W_{ia}x_i$	Matrix-vector product $W^\top x$	Column currents in a crossbar
Hidden activation $m_a = \sigma(\beta I_a)$ or softmax	Nonlinear hidden response	Comparator/sigmoid/softmax circuit, stochastic device, or oscillator nonlinearity
Visible feedback $\sum_a W_{ia}m_a$	Matrix-vector product Wm	Row currents using same/tied crossbar or reciprocal array
Score/force $f_i = -\partial_i \mathcal{F}$	Visible force	Current injection into visible oscillator node
Column aggregate $W_{:a}^\top \delta$	Error projection	Crossbar column read with error voltages
Local update ΔW_{ia}	Conductance programming	Coincident row/column pulses plus modulatory gates

This is a strong architectural match. Dense visible-visible all-to-all interactions are difficult to wire directly; a visible-hidden crossbar is a natural physical primitive.

10.2 Implementation constraints

The main constraints are:

- **Reciprocity/tied weights.** The energy formulation assumes that the forward hidden field and backward visible force use the same W . Crossbars can be read in both directions in principle, but peripheral asymmetry and device mismatch must be calibrated. If two arrays are used, tying must be enforced or learned around.
- **Signed couplings.** Conductances are nonnegative. Use differential pairs, offset encodings, or four-quadrant transconductance cells.
- **Hidden temperature.** The free-energy identities assume a known β . A deterministic sigmoid hidden circuit can implement the mean m_a , but the susceptibility χ_a must match the effective temperature used in the score.
- **Time-dependent protocol.** Diffusion needs $W(t)$, $b(t)$, or gains that change with reverse time. Reprogramming a full crossbar at every time step is unattractive. Prefer a small number of basis arrays with time-dependent analog gains, or a fixed W with time-dependent hidden biases/temperatures.
- **Update circuitry.** The exact rule is not a pure two-factor Hebbian update. It is a three-factor or four-factor local rule involving visible clamp x_i , hidden activation/susceptibility, and visible score error or force. This is still local/column-local, but it requires peripheral support.
- **Finite time-scale separation.** If τ_h is not much smaller than the visible dynamics, hidden variables do not represent an instantaneous free energy. The visible process becomes non-Markovian with extra lag/friction.

10.3 Why restricted is better than deep/hierarchical for a first paper

A deep or hierarchical latent DAM would be more expressive, but it weakens the physical and mathematical story. The posterior $q(h|x)$ becomes harder to equilibrate and harder to differentiate. Prior work on

general latent energy-based score matching often requires variational inference or bilevel optimization. A one-hidden-layer restricted model has tractable conditionals, local hidden statistics, and a direct crossbar implementation. It is the right first proposal.

11 Novelty and relation to prior work

11.1 Closest existing pieces

The closest relevant lines of work are:

- **Diffusion as associative memory / DenseAM.** The recent DenseAM-diffusion literature shows that diffusion energies can be interpreted as modern Hopfield/DenseAM energies and that memorization, spurious states, and generalization can be studied as associative-memory phases.
- **Score matching for latent energy-based models.** Gaussian-binary RBMs and related latent EBMs have been trained with score matching, and more general latent EBMs have been approached with bilevel score matching.
- **Physical/local learning for score-based models.** The visible-only oscillator work establishes that score-based generative driving protocols can be trained from local force measurements or short-time dynamics.
- **Analog/crossbar associative memory.** Recent work has argued that DenseAMs and high-capacity associative memories can be implemented in analog circuits and memristor crossbars.

The proposed contribution would be at the intersection: a marginal free-energy, hidden-unit extension of local physical force matching, designed from the beginning for crossbar implementation and score-based reverse diffusion.

11.2 Novelty judgment

My novelty judgment is: **moderately high as a synthesis, not high as a set of isolated mathematical components.** A strong paper should not claim that latent score matching, RBMs, DenseAM, or crossbar associative memories are new. It should claim a specific new combination:

A fast-equilibrating restricted DenseAM/RBM hidden layer can be adiabatically eliminated to yield a visible free energy whose score is trainable by local on-device force matching under full-amplitude visible clamps, giving a physically implementable latent score-based generative model in crossbar-coupled oscillator hardware.

That claim appears defensible and interesting. It would be especially strong if accompanied by either (i) a hardware-aware simulation with quantized/asymmetric crossbar nonidealities, or (ii) a small physical prototype or SPICE-level circuit simulation showing that the hidden equilibration and update measurements are feasible.

12 Risks and failure modes

1. **Hidden equilibration may be too slow.** If hidden units do not relax before visible forces are sampled, the learned rule no longer matches the free-energy gradient. This can be tested by sweeping τ_h/τ_x .
2. **Susceptibility measurements may dominate complexity.** The exact rule needs χ_a or categorical covariance. A deterministic hidden activation circuit can expose this analytically, but stochastic physical hidden units may require repeated samples.

3. **The model may memorize.** Categorical DenseAM with one hidden unit per pattern can reproduce empirical memories. Generalization requires capacity control, temperature/noise scheduling, prototype compression, sparsity, or random-feature memory compression.
4. **Weight tying may break.** Nonreciprocal crossbar readout breaks the energy interpretation and can create nonconservative score fields. This may still generate samples, but it loses the clean DAM/free-energy argument.
5. **Time-dependent crossbar programming may be impractical.** A continuously varying $W(t)$ is expensive. Use time-basis factorization, time-varying gains, or train a static hidden dictionary with time-varying biases.
6. **Locality is not purely synapse-local.** The exact update is local plus row/column broadcasts. This is acceptable for crossbar hardware, but the claim should be phrased as local/column-local rather than purely two-terminal local.

13 Experiments that would make the proposal convincing

13.1 Numerical gradient check

Before any generative experiment, verify the rule. For a small binary hidden model, compare Eq. (38) and Eq. (50) against automatic differentiation of the explicit free energy. This validates signs, susceptibility terms, and temperature factors.

13.2 Two-dimensional toy problem

Use a ring, two moons, or a mixture of Gaussians. Compare:

1. visible-only polynomial oscillator energy;
2. binary hidden free-energy model;
3. categorical DenseAM/prototype model;
4. exact neural score network baseline.

Measure mode coverage, energy landscape, learned score error, and sensitivity to hidden time-scale ratio.

13.3 MNIST 0/1 continuation of the oscillator work

Use the existing 12×12 downsampled MNIST setup. Add hidden units $m \in \{16, 32, 64, 128, 256\}$. Evaluate:

- sample quality and diversity;
- nearest-neighbor memorization;
- spurious-state rate using the DenseAM/diffusion diagnostics;
- robustness to limiting visible-hidden connection range versus full crossbar;
- crossbar quantization and stuck-device noise.

The hypothesis is that hidden units should improve global digit coherence with fewer visible-visible long-range couplings.

13.4 Hardware-aware ablations

Simulate differential-pair conductances, finite precision, asymmetric forward/backward arrays, stochastic hidden activations, and finite hidden equilibration time. The proposal becomes substantially stronger if the rule remains stable under realistic nonidealities.

14 Suggested paper framing

A concise framing would be:

We extend local force matching for physical score-based generative models from visible-only oscillator energies to latent free energies. Hidden units equilibrate rapidly under a full-amplitude visible clamp, producing a marginal score through adiabatic elimination. For restricted DenseAM/RBM architectures, the resulting denoising force-matching rule depends only on local visible states, hidden activations/susceptibilities, measured visible forces, and row/column crossbar aggregates. This yields a physically implementable path to more expressive on-device score models without infinitesimal nudging or negative-phase equilibrium sampling.

The headline should not be “deep latent diffusion in hardware.” The stronger and cleaner headline is “fast latent DenseAM free energies for local physical score matching.”

15 Conclusion

The proposal is worth pursuing. The most important conceptual move is to replace the visible-only learned energy by a hidden-marginal free energy. This directly connects DenseAM log-sum-exp energies, RBM hidden-unit expressivity, and physical force matching. A restricted visible-hidden architecture is not just a hardware compromise; it is the architecture in which the math stays local and the physical implementation remains plausible.

The main recommended research path is:

1. derive and validate the free-energy denoising rule for a one-hidden-layer restricted model;
2. implement it in simulation on the same oscillator benchmarks as the visible-only work;
3. show that hidden units improve expressivity/sample quality at fixed visible coupling complexity;
4. demonstrate that the update is robust to crossbar-like constraints;
5. only then explore deeper or more hierarchical hidden structures.

If the experiments show a clear quality/capacity gain without sacrificing the full-amplitude clamp and local update story, the idea could form a compelling follow-up paper.

A Derivation of the free-energy identities

Let

$$Z_h(x) = \int e^{-\beta E(x,h)} dh, \quad \mathcal{F}(x) = -\beta^{-1} \log Z_h(x). \quad (54)$$

Then

$$\partial_i \mathcal{F} = -\beta^{-1} \frac{1}{Z_h} \partial_i Z_h = -\beta^{-1} \frac{1}{Z_h} \int -\beta (\partial_i E) e^{-\beta E} dh = \langle \partial_i E \rangle_q. \quad (55)$$

For any statistic $A(x, h)$,

$$\partial_i \langle A \rangle_q = \langle \partial_i A \rangle_q - \beta \text{Cov}_q(A, \partial_i E). \quad (56)$$

Setting $A = \partial_j E$ gives the Hessian identity

$$\partial_i \partial_j \mathcal{F} = \langle \partial_i \partial_j E \rangle_q - \beta \text{Cov}_q(\partial_i E, \partial_j E). \quad (57)$$

Setting $A = \partial_\theta E$ gives the mixed derivative identity

$$\partial_i \partial_\theta \mathcal{F} = \langle \partial_i \partial_\theta E \rangle_q - \beta \text{Cov}_q(\partial_\theta E, \partial_i E). \quad (58)$$

These identities are the mathematical basis for measuring hidden means and fluctuations under a visible clamp.

B Zero-temperature hidden units

If hidden equilibration is deterministic and effectively zero-temperature, then

$$\mathcal{F}_0(x) = \min_h E(x, h). \quad (59)$$

When the minimizer $h^*(x)$ is unique and smooth, the envelope theorem gives

$$\nabla_x \mathcal{F}_0(x) = \nabla_x E(x, h^*(x)). \quad (60)$$

This is attractive for hardware because only the hidden equilibrium state is needed for the force. However, the Hessian and the exact score-matching gradient include derivatives of $h^*(x)$ with respect to x , and nonsmooth switching between hidden states can be problematic. A finite-temperature or smooth deterministic hidden activation is safer for score-based training.

C Reference notes used in the report

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